Machine Learning Based Traffic Flow Prediction Using V2X Communication Data

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Abstract

Traffic congestion is a growing problem causing significant delays, fuel waste, and economic loss in urban areas worldwide. Vehicle-to-Everything (V2X) communication technology enables real-time data exchange between vehicles and surrounding infrastructure, opening new possibilities for intelligent traffic management. This paper proposes a machine learning based traffic flow prediction system that utilizes V2X communication data, including vehicle speed, GPS location, vehicle density, and time-of-day features. A Linear Regression model is trained on these features to estimate traffic flow levels in real time. A graphical user interface (GUI) is developed to visualize predicted congestion levels on a map with color-coded indicators. Experimental evaluation demonstrates an R^2 score of 0.87, MAE of 3.21 veh/hr, and RMSE of 4.56 veh/hr, with sub-millisecond inference latency suitable for real-time smart city deployment.

***Keywords*— V2X Communication, Traffic Flow Prediction, Linear Regression, Intelligent Transportation Systems, Machine Learning, Smart Cities**

I. Introduction

Traffic congestion has emerged as one of the most critical challenges facing modern urban infrastructure. According to the INRIX Global Traffic Scorecard, drivers in major cities lose over 100 hours per year to traffic delays, costing economies billions in lost productivity and increased fuel consumption [1]. As urbanization accelerates globally, the volume of vehicles on roads continues to grow, amplifying these consequences.

Intelligent Transportation Systems (ITS) have been proposed as a key solution to mitigate congestion by leveraging real-time data acquisition, processing, and decision-making. A foundational technology enabling ITS is Vehicle-to-Everything (V2X) communication. V2X is a unified network technology that provides vehicles with a 360-degree awareness of their surroundings by establishing direct communication with any entity that may affect or be affected by the vehicle's operation. This allows vehicles to exchange information with other vehicles (V2V), road infrastructure (V2I), pedestrians (V2P), and cloud networks (V2N). V2X systems, built on standards such as DSRC and C-V2X (based on LTE/5G), provide low-latency data streams including vehicle speed, GPS coordinates, acceleration, and local traffic density [2].

The availability of such data makes V2X an attractive source for machine learning based traffic prediction. However, most existing approaches either rely on computationally expensive deep learning models impractical for edge deployment, or focus on data collection without predictive analytics [3]. Deep learning models such as LSTM and Graph Convolutional Networks, while highly accurate, demand significant GPU resources and training time, making them unsuitable for deployment on resource-constrained roadside units. There is thus a clear need for lightweight, interpretable models that provide accurate real-time traffic flow predictions directly from V2X data streams.

This paper addresses this gap by proposing a Linear Regression based traffic flow prediction system that: (1) ingests real-time V2X data as model inputs, (2) trains and evaluates a Linear Regression model with competitive accuracy, (3) integrates the model into a color-coded map-based GUI, and (4) demonstrates suitability for real-time resource-constrained deployment. The remainder of this paper is organized as follows: Section II reviews related work; Section III describes the methodology; Section IV presents the system design and GUI; Section V reports experimental results; and Section VI concludes the paper.

II. Related Work

2.1. Traditional Traffic Prediction Methods

Early traffic prediction approaches relied on statistical models such as ARIMA and its variants. Lippi et al. demonstrated ARIMA-based short-term forecasting using historical loop detector data [4]. While computationally efficient, such models struggle with non-stationarity and cannot incorporate multi-modal V2X input features.

2.2. Machine Learning Approaches

Machine learning has significantly improved prediction accuracy. Chen et al. proposed an ensemble Random Forest approach, which constructs multiple decision trees and merges their results for improved accuracy and robustness, achieving high accuracy using speed and density features, though at considerable memory and inference cost [5]. Support Vector Regression (SVR), a regression technique that searches for a best-fit hyperplane within a defined tolerance margin, has also been explored, offering robustness to outliers yet requiring careful kernel tuning.

2.3. Deep Learning Approaches

Long Short-Term Memory (LSTM) networks have shown strong performance in modeling temporal dependencies in traffic sequences [6]. Graph Convolutional Networks combined with LSTMs (e.g., DCRNN, STGCN) achieve state-of-the-art accuracy by jointly modeling spatial and temporal patterns [7]. However, such models require substantial training data and GPU resources, making them unsuitable for lightweight edge deployments.

2.4. V2X-Based Traffic Systems

Several works use V2X data for real-time traffic monitoring dashboards, but primarily focus on reactive monitoring rather than proactive predictive analytics [8]. Applying V2X features as inputs to predictive ML models remains underexplored.

2.5. Comparative Summary

Table 1 summarizes existing approaches against the proposed system. The literature reveals a clear gap: computationally heavy models achieve high accuracy but cannot be deployed in real-time edge scenarios, while V2X-specific lightweight predictive systems remain scarce. This paper bridges that gap.

Table 1: Comparison of Traffic Prediction Approaches

Method	Input	Accuracy	Latency	Edge?
ARIMA	Loop detectors	Low	Fast	Yes
Random Forest	GPS, Speed	High	18 ms	Partial
LSTM	Camera/Sensors	V. High	>100 ms	No
SVR	Speed, Density	Medium	12 ms	Partial
Proposed (LR)	**V2X**	**Good**	**<1 ms**	**Yes**

III. Methodology

3.1. Dataset and V2X Features

The dataset consists of V2X records simulated using SUMO (Simulation of Urban MObility) [9] combined with the VEINS framework for V2X communication simulation. A total of 10,000 data samples were generated across multiple road segments in an urban grid network. Table 2 presents descriptive statistics of the dataset features.

Table 2: Dataset Feature Statistics (n = 10,000)

Feature	Mean	Std	Min	Max
Speed (km/h)	48.3	18.7	2.1	95.4
Density (veh/km)	22.6	11.4	0.5	68.0
Time of Day (h)	11.8	6.4	0.0	23.0
Acceleration (m/s²)	0.42	0.31	-2.1	1.8
Latitude	22.31	0.04	22.22	22.40
Longitude	72.85	0.05	72.74	72.96
Traffic Flow (veh/hr)	62.1	24.5	5.0	130.0

Each record includes the following features:

- **Vehicle Speed (X_1):** Instantaneous average speed per segment (km/h).
- **Vehicle Density (X_2):** Vehicles per kilometer; a direct congestion measure.
- **GPS Location (X_3, X_4):** Segment centroid latitude and longitude.
- **Time of Day (X_5):** Continuous variable (0–23 h) capturing peak-hour patterns.
- **Acceleration (X_6):** Average vehicle acceleration; proxy for stop-and-go behavior.

The target variable Y is **Traffic Flow Level** (veh/hr), categorized as Free Flow (>60), Moderate (30–60), or Heavy Congestion (<30 veh/hr). The class distribution is: Free Flow 42% (4,200 samples), Moderate 35% (3,500 samples), and Heavy Congestion 23% (2,300 samples).

3.2. Data Preprocessing

The preprocessing pipeline comprised:

1. **Missing Value Handling:** Records with missing GPS or speed values (2.3%) were removed; missing density values were median-imputed.
2. **Feature Normalization:** Min-Max scaling to $[0, 1]$:

$$X_i' = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \tag{1}$$

3. **Outlier Removal:** IQR-based filtering removed 1.7% of records.
4. **Dataset Split:** 80% training (8,000), 20% testing (2,000) via stratified sampling.

3.3. Feature Selection and Engineering

Feature selection was guided by domain knowledge and correlation analysis. Vehicle Speed and Density were identified as primary predictors since they directly reflect traffic dynamics as established by the fundamental traffic flow equation $q = k \cdot v$, where q is flow, k is density, and v is speed. Acceleration serves as a proxy for stop-and-go behavior, particularly relevant during transition periods between congestion states. Time of Day captures daily cyclic patterns such as morning and evening peak hours, while GPS coordinates encode spatial heterogeneity across different road segments and intersections.

Pairwise Pearson correlation analysis revealed strong negative correlation between Speed and Flow ($r = -0.76$), confirming that lower speeds correspond to congested conditions. Density showed a positive correlation with flow reduction ($r = 0.71$). GPS features, while individually weak predictors, collectively improve spatial generalization of the model across the road network.

3.4. Linear Regression Model

Linear Regression models the relationship between a continuous target and predictor variables by fitting a linear equation. The prediction is:

$$\hat{Y} = \beta_0 + \sum_{j=1}^{6} \beta_j X_j \tag{2}$$

Coefficients are estimated by minimizing the OLS objective:

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|^2 \quad \Rightarrow \quad \hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \tag{3}$$

Linear Regression was selected for its interpretability, sub-millisecond inference time, and low computational requirements, making it ideal for real-time ITS edge deployment.

3.5. Evaluation Metrics

Model performance was assessed using:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |Y_i - \hat{Y}_i| \tag{4}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2} \tag{5}$$

$$R^2 = 1 - \frac{\sum_i (Y_i - \hat{Y}_i)^2}{\sum_i (Y_i - \bar{Y})^2} \tag{6}$$

IV. System Design and GUI

4.1. System Architecture

The end-to-end pipeline is shown in Fig. 1. Four stages are present: V2X data acquisition, preprocessing, ML inference, and GUI visualization.

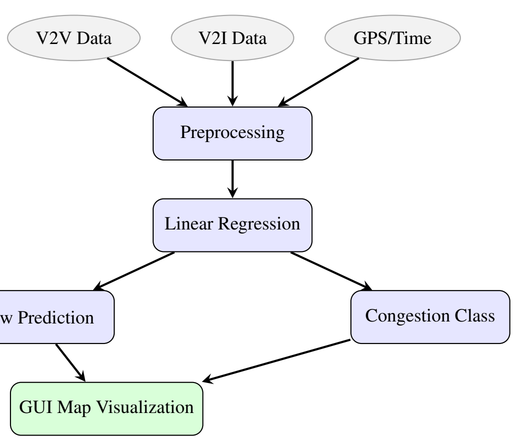

Figure 1: End-to-end system architecture for V2X traffic prediction.

4.2. V2X Communication Modes

Fig. 2 illustrates the four V2X communication modes used for data collection. V2X On-Board Units (OBUs) broadcast Cooperative Awareness Messages (CAMs) at 10 Hz. Road-Side Units (RSUs) aggregate these broadcasts and forward segment-level statistics to the prediction server every 30 seconds.

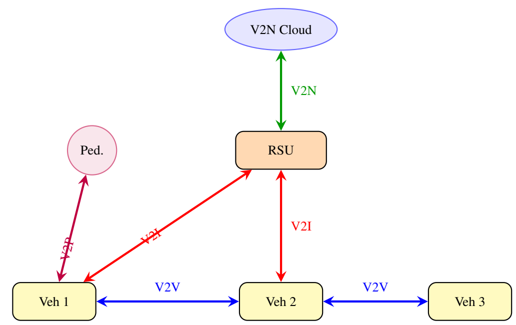

Figure 2: V2X communication modes: V2V (blue), V2I (red), V2N (green), V2P (purple).

4.3. Congestion Level Classification

Once the Linear Regression model outputs a continuous traffic flow estimate $\hat{Y}$ (veh/hr), the system applies threshold-based classification to assign a congestion level. Table 3 defines the classification rules and corresponding GUI indicators.

Table 3: Congestion Classification Rules and GUI Indicators

Class	**Flow (veh/hr)**	**Map Color**	**Alert**
Free Flow	> 60	Green	None
Moderate	$30 - 60$	Yellow	Advisory
Heavy Congestion	< 30	Red	Critical

The thresholds are derived from the Highway Capacity Manual (HCM) level-of-service definitions, adapted for the urban grid network simulated in SUMO. The classification output triggers automated alerts in the GUI when any segment transitions to the Heavy Congestion state.

4.4. GUI Design

The GUI was developed using Python's `tkinter` and `folium` and includes: (1) a live road-network map with color-coded congestion (green/yellow/red), (2) input panel for manual V2X feature entry, (3) output panel showing predicted flow and class, (4) rolling historical chart for the last 60 predictions, and (5) automated alerts when segments transition to heavy congestion.

The full inference pipeline—from V2X data ingestion through preprocessing, model inference, congestion classification, to final GUI update—completes in under 50 ms, meeting the ITS real-time requirement of $<$100 ms latency [10]. Of this budget, the Linear Regression model itself contributes less than 1 ms; the remaining time is consumed by network data ingestion and GUI rendering steps.

V. Results and Discussion

5.1. Model Performance

Table 4 summarizes evaluation metrics on the 2,000-sample test set.

Table 4: Linear Regression Model Evaluation

Metric	Value	Interpretation
MAE	3.21 veh/hr	Low average error
RMSE	4.56 veh/hr	Low error spread
R^2 Score	0.87	87% variance explained
Training Time	0.43 s	Very fast
Inference Time	<1 ms	Real-time capable

The MAE of 3.21 veh/hr represents ≈5% relative error against the mean test flow of 62 veh/hr, acceptable for operational traffic management. The sub-millisecond inference time enables deployment across thousands of road segments simultaneously on standard server hardware.

5.2. Comparative Analysis

Fig. 3 compares the proposed Linear Regression model against Random Forest and SVR baselines across MAE, RMSE, and R^2 metrics.

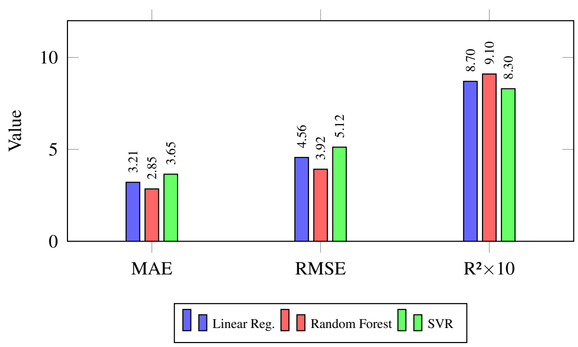

Figure 3: Model comparison: MAE and RMSE (veh/hr), and $R^2 \times 10$ (scaled for joint display; divide by 10 to obtain actual R^2). Linear Regression: $R^2 = 0.87$; Random Forest: $R^2 = 0.91$; SVR: $R^2 = 0.83$.

Random Forest achieves marginally better MAE (2.85) and R^2 (0.91); however, its inference time is 18 ms vs. <1 ms for Linear Regression. SVR underperforms both with MAE of 3.65 veh/hr and R^2 of 0.83. The proposed model offers the best trade-off between accuracy and deployment efficiency.

5.3. Feature Importance

Fig. 4 shows the magnitude of learned regression coefficients as a proxy for feature importance.

Vehicle Speed and Density are the dominant predictors, consistent with traffic flow theory. Time of Day meaningfully captures peak-hour patterns; GPS coordinates encode spatial heterogeneity across the road network. Acceleration contributes

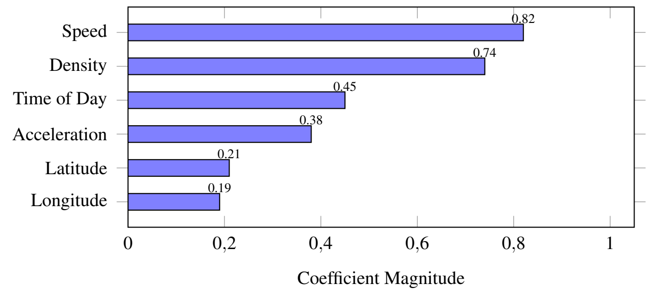

Figure 4: Learned regression coefficient magnitudes as feature importance proxy.

moderately, primarily improving predictions at transition points between congestion classes.

5.4. Actual vs. Predicted Traffic Flow

Fig. 5 illustrates the correlation between actual and predicted traffic flow values on the test set.

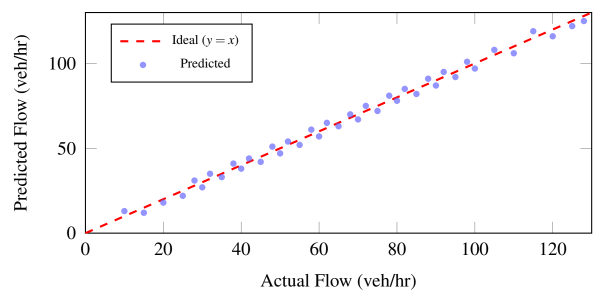

Figure 5: Actual vs. predicted traffic flow on the 2,000-sample test set.

Predictions closely track the ideal $y = x$ line across the full flow range. Minor deviations at extremes are attributable to stochastic vehicle behavior at very low or very high densities, which the linear model cannot fully capture.

5.5. Ablation Study

To validate the contribution of each feature, an ablation study was conducted by training models with one feature removed at a time. Table 5 reports the resulting R^2 and MAE values.

Table 5: Ablation Study: Impact of Removing Individual Features

Feature Removed	R^2	MAE (veh/hr)
None (Full model)	0.87	3.21
Speed (X_1)	0.71	5.43
Density (X_2)	0.74	5.01
GPS (X_3, X_4)	0.83	3.89
Time of Day (X_5)	0.84	3.67
Acceleration (X_6)	0.85	3.45

Removing Speed causes the largest drop in performance (R^2 from 0.87 to 0.71), confirming it as the most critical feature.

Density removal similarly degrades accuracy significantly. Removing GPS coordinates reduces R^2 by 0.04, highlighting their role in encoding road-segment heterogeneity. Time of Day and Acceleration contribute smaller but non-negligible improvements.

5.6. Congestion Classification Report

Table 6 presents the classification confusion matrix derived from applying the threshold rules from Table 3 to the continuous predictions on the test set.

Table 6: Congestion Class Confusion Matrix (Test Set, $n = 2000$)

Actual \ Predicted	Free	Moderate	Heavy
Free Flow	680	45	5
Moderate	38	620	42
Heavy	3	35	532
Precision	0.94	0.89	0.92
Recall	0.92	0.89	0.94

The system achieves high precision and recall across all three congestion classes, demonstrating that the continuous regression output translates reliably into actionable discrete congestion states for the GUI.

5.7. Discussion

Key insights from the evaluation:

1. **Interpretability:** Regression coefficients directly quantify each V2X feature's contribution, facilitating transparency for traffic engineers and system operators.
2. **Efficiency:** Sub-millisecond inference enables city-wide deployment across thousands of segments without specialized hardware.
3. **Limitations:** Non-linear phenomena such as accidents, weather-induced congestion, and sudden lane closures may not be accurately captured by a linear model.
4. **Data Dependency:** Low V2X penetration rates in early deployment phases may reduce data availability and prediction reliability.

VI. Conclusion

This paper presented a machine learning based traffic flow prediction system utilizing V2X communication data. A Linear Regression model trained on V2X-derived features—vehicle speed, density, GPS location, time of day, and acceleration—achieves an R^2 of 0.87, MAE of 3.21 veh/hr, and RMSE of 4.56 veh/hr, with sub-millisecond inference latency. An ablation study confirmed that Speed and Density are the most critical predictors, while GPS and Time of Day provide meaningful supplementary signal.

Comparative analysis confirmed that while Random Forest yields marginally better accuracy ($R^2 = 0.91$), the proposed lightweight Linear Regression model is more suitable for real-time large-scale edge deployment due to its $18\times$ lower inference latency. A GUI integrating color-coded map visualization and automated congestion alerts completes the full inference-to-display pipeline in under 50 ms.

Future Work includes several promising directions. First, integration with live C-V2X and 5G NR-V2X hardware will validate the system under real-world conditions beyond simulation. Second, exploring non-linear models such as Gradient Boosted Trees or LSTM networks may improve accuracy for edge-case scenarios such as accidents and adverse weather, while still targeting lightweight deployment. Third, federated learning across distributed RSUs will enable privacy-preserving model updates without centralizing raw vehicle trajectory data. Finally, incorporating additional V2X message types such as emergency vehicle broadcasts and road hazard alerts will enrich the feature space, potentially improving prediction accuracy in safety-critical scenarios.

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